

Response for LW18078

Dear Reviewers and Editor,

we thank both Reviewers for their valuable suggestions and constructive comments.

In the report both Reviewers have kindly appreciated our manuscript as “well-written and easy to follow” as well as “written in a very clear and well structured way”. Specifically the Reviewer B has highly recommended our manuscript for publication as “I am impressed by the results and believe that the manuscript can be well suited for publication in Physical Review Letters”, while the Reviewer A commented our work positively with “both the experimental and theoretical foundations to enable this new type of PEEM imaging” but has raised several detailed physical as well as technical questions. These issues are clarified point-by-point as in the following. With our modifications and our response, we are looking forward to publication of the manuscript in Physical Review Letters.

Reviewer A

(A.1) Reviewer A: “However, there are several deficiencies that I recommend to address before it can be considered for publication in a peer-reviewed journal. Even if those deficiencies are appropriately addressed, I am not convinced that it would qualify for PRL in 2024 given a lack of novelty and impact. It might be more suitable in PRB, particularly since this would also allow for avoiding any SI and moving all relevant information to the main section.

Reply: Our work is the first dark-field photoelectron microscopy (PEEM) on magnetic materials worldwide. As highlighted in the introduction of our manuscript, our joint experimental and theoretical results here represent a significant breakthrough based on the technical development over the last two decades since G. K. L. Marx *et al.*, Phys. Rev. Lett. **84**, 5888 (2000). In view of our detailed reply below addressing all the questions from both Reviewers, we hope that our choice of the Physical Review Letters can be justified. The supplement provides only the background and the details of the theoretical code used. All relevant information is included in the main manuscript.

(A.2) Reviewer A: "Setting a broader context for this work: There are many other microscopy techniques being developed / available these days that would allow ultrafast spin dynamics studies. I suggest mentioning ultrafast STM, particularly in view of the surface system studied here, ultrafast LTEM studies, which also use lab lasers but will provide far better spatial resolution.

Reply: We thank the Reviewer for this valuable suggestion, which significantly broadens the readership of our manuscript. Despite that the ultrafast scanning tunneling microscopy (STM) could offer the unprecedented spatial resolution down to the single atomic scale, its access to electron spin at surfaces relies on the spin-polarized current from the tip (Nat. Nano. **9**, 588 (2014)). Therefore the recent development of ultrafast STM using THz electric field (Prog. Surf. Sci. **99**, 100727 (2024)) may only provide the spin information if spin-orbit split electronic states can be probed in the tunneling process (Nat. Photonics. **18**, 595 (2024)). In contrast, the combination of ultrafast lasers with the Lorentz transmission electron microscopy (LTEM) can offer the direct magnetic contrast on the femtosecond time domain (Phys. Rev. X **8**, 031052 (2018); Commun. Phys. **3**, 36 (2020); Phys. Rev. X **12**, 041030 (2022)). The information above is now added in the introduction paragraph of our manuscript.

(A.3) Reviewer A: "With regard to PEEM, I suggest mentioning XFEL attempts for PEEM and compare to their darkfield technique. What about charging effects?"

Reply: Photoemission microscopy (PEEM) can be efficiently combined with the x-ray free electron laser (XFEL) as has been nicely demonstrated in the literature (Rev. Sci. Instrum. **91**, 013109 (2020)). Due to the low pulse repetition rates of the XFEL, space-charge effects can occur and additional measures need to be taken for the PEEM experiments (Rev. Sci. Instrum. **92**, 053703 (2021); New J. Phys. **20**, 033004 (2018)). In strong contrast to the low repetition rates of XFEL, we have combined PEEM with the megahertz repetition rates of our high-power laser system as well as the Hg discharge lamp, thereby minimizing the space-charge effects. The references above are now added in the introduction of the manuscript.

(A.4) Reviewer A: "How does this work relate to momentum microscopy, which is currently a hot topic?"

Reply: As has been reviewed in detail by the early publication of the momentum microscopy (Rev. Sci. Instrum. **79**, 053702 (2008)), its original development stems from the back focal

plane imaging in a photoelectron microscope (PEEM) (Rev. Sci. Instrum. **74**, 2754 (2003)). Therefore, our approach can be directly implemented into the momentum microscopy. This information is now mentioned in the outlook at the end of our manuscript.

(A.5) Reviewer A: "Intensity and contrast are one aspect, but for microscopy spatial resolution is inevitable. Fig. 4 seems to cover a field-of-view (FOV) of more than 50 μm , from which I would derive a medium spatial resolution. How does that compare to standard magneto-optical Kerr effect (MOKE)?"

Reply: As formulated in the standard literature for the cathode lens in the photoelectron microscope (PEEM) with great details, such as the equation (7) in Ultramicroscopy **27**, 273 (1989), the selection of the central region of the back focal plane imaging with an aperture increases the spatial resolution by reducing the spherical aberration. Our darkfield imaging method corresponds to the same selection but with a small offset angle from the central axis of the electron optics. Therefore the spatial resolution would be slightly degraded from the optimal spatial resolution of our PEEM of about 40 nm depending on the exact size of the selected parallel momentum region (J. Phys.: Condens. Matter **11**, 9517 (1999)). As an order-of-magnitude estimation using the quadratic angular dependence of the aberration in the equation (7) above, with the selection of about $\pm 1/6$ of the whole momentum region and an offset of $1/3$ of the whole momentum region, the off-center electron ray has a maximal angle which is increased by about a factor of 3 from that of the central momentum selection. These conditions are comparable to our experimental settings in Fig. 4. As a result of the quadratic angular dependence of the spherical aberration, the spatial resolution would be reduced from 40 nm to about 360 nm, and this estimation is in reasonable agreement with our data in Fig. 4. For comparison, the expected domain walls here have a width of about 200 nm (Phys. Rev. Lett. **62**, 819 (1989)). Note that the ultimate resolution of PEEM-based magnetic imaging depends on the initial resolution of the instrument (in a standard mode) as well as the reduction due to the off-center electron detection. The latter is unavoidable for darkfield imaging and deteriorates the resolution by a factor between three to nine, depending on the angular photoelectron distribution. The information above is now added in the last discussion paragraph of the manuscript.

The magneto optical Kerr effect (MOKE) can be implemented with standard optical microscopy, and its ultimate spatial resolution can reach 10 nm in combination with near-field optics (Rev. Sci. Instrum. **67**, 715 (1996)). For the time-resolved magnetic imaging in combination with ultrafast lasers, a spatial resolution of about several hundreds of nanometers

has been demonstrated as given by the diffraction limit and the wavelength of the femtosecond laser (Rev. Sci. Instrum. **88**, 123708 (2017)).

We added the information on the resolution of the magneto-optical Kerr Effect in the introduction of the manuscript.

(A.6) Reviewer A: "The authors focus on in-plane systems, which is most likely true for their thin film, but the community is currently excited about orientations that are neither fully in-plane nor out-of-plane. Can this technique distinguish those arrangements, e.g. by varying the geometry?"

Reply: As demonstrated in our manuscript in detail, all the possible in-plane domain orientations can be fully resolved by exactly the same number of possible momentum selections at the back focal plane (Fig. 4). In addition, symmetric momentum selection around normal emission is sensitive to the out-of-plane orientation of the magnetization as shown in Phys. Rev. Lett. **96**, 237402 (2006). Therefore, one is able to address all three different magnetization axes. However, the full 3D reconstruction of the local magnetization direction is beyond the present work.

For more delicate local non-collinear magnetization orientations such as vortex structures, further experimental degrees of freedom will be required to disentangle these orientations. Such additional degrees of freedom, for example, can include the polarization state of light, the spatial phase structure of the light wave-front, the time delay between successive light pulses, as well as the sample orientation. More specifically, the complementary combination of magnetic dichroic signals with linearly and circularly polarized light (Adv. Electron. Mater. **8**, 2200162 (2022)), the orbital angular momentum of light wave-front (Phys. Rev. B **107**, L060407 (2023)), the time-dependent interference in the nonlinear photoemission processes (Phys. Rev. Research **6**, 013163 (2024)), as well as the consecutive measurements with sample rotation (J. Synchrotron Rad. **31**, 336 (2024)) have been demonstrated to image the more delicate local non-collinear magnetization orientations.

(A.7) Reviewer A: "Symmetric exchange interaction is the dominant interaction, what about the antisymmetric exchange interaction (DMI)?"

Reply: The magnetic dichroism reported in our manuscript results from the presence of both spin-orbit interaction and exchange interaction in the electronic structure of bulk Fe. Both of these spin-dependent interactions are also involved in the origin of the Dzyaloshinskii-Moriya interaction (DMI, I. Dzyaloshinskii, J. Phys. Chem. Sol. **4**, 241 (1958); T. Moriya, Phys. Rev.

120, 91 (1960)). However, as has been nicely formulated by T. Moriya in his publication above, the DMI is a direct extension of the super-exchange interaction as modelled by P. W. Anderson, which mainly describes antiferromagnetic transition metal compounds with low electrical conductivity (Phys. Rev. **115**, 2 (1959)). Despite that the DMI would hardly be applicable for the bulk Fe in our manuscript, it could be important for ultrathin Fe films grown on heavy metal substrates (Phys. Rev. Lett. **104**, 137203 (2010)). In addition, the DMI in transition metal compounds (Proc. Phys. Soc. **79**, 394 (1962)) has been previously observed by magnetic dichroic PEEM (Phys. Rev. Lett. **106**, 107201 (2011)). We would like to add that for the basic analysis, only symmetry considerations are needed, the origin of the magnetic texture - Heisenberg interaction or DMI - is irrelevant.

(A.8) Reviewer A: "I am confused about Figs. 2 and 3 when comparing experiments with theory. The color codes seem to indicate a difference of about a factor 2 (40 vs 20 and 10 vs 6), but why would then the difference between max and min not be visible in the experimental data? They seem to be much weaker in contrast. Is there a mistake in the display settings? Also, why are the experimental data so noisy? Is this a result of statistics? If so, why not accumulate better statistics?"

Reply: We thank the Reviewer for carefully pointing out that there exists a factor of 2 to 3 difference between the absolute values of the dichroic asymmetries A_{pol} (Fig. 2) and A_{ex} (Fig. 3) from the experiments and theory. In our calculations, we did not use any specific *ad hoc* parameters in order to remove this discrepancy between the experiments and theory. Furthermore, for our experimental data in these figures, the difference between the maximum and minimum in the A_{pol} and A_{ex} can be identified by looking at the most intense red and blue regions in the left columns of Fig. 2 and Fig. 3. For a fair comparison, in these figures we specifically fix the range of color coding, so that the data at different energies can be directly compared. Our calculations describe the direct photoemission process but do not include secondary electrons due to scattering events. The latter are always present in experimental data and reduce the absolute value of the dichroic asymmetries.

The fluctuation in our experimental data mainly reflects the counting statistics of photoelectrons, which could be reduced if an order-of-magnitude longer measurement time would be applied. However, the improvement of the counting statistics at small binding energies close to the Fermi level (E_F), such as A_{ex} at 0.05 eV in the top-most left panel of Fig. 3, would still be limited intrinsically by the reduced photoemission intensity close to E_F . More importantly, the main physics discussed in our manuscript will still hold as supported by

the reasonable qualitative agreement between our experiments and theoretical calculations in Fig. 2 and Fig. 3.

(A.9) Reviewer A: "Fig. 4 needs a length scale bar and I highly encourage the authors to discuss spatial resolution."

Reply: The size of all figures in Fig. 4 is $56\ \mu\text{m} \times 56\ \mu\text{m}$ as written in the figure caption, and the spatial resolution is addressed above in **(A.5)** in great detail as well as added in the manuscript.

(A.10) Reviewer A: "ARPES in Ref. [25, 28]. Why do the authors only mention their own work? There are numerous ARPES reviews, which might be better suited."

Reply: We thank the Reviewer for this hint, but in that paragraph of our manuscript we are only citing our experimental setup (to shorten the manuscript) and do not discuss any ARPES spectra.

(A.11) Reviewer A: "Last sentence: "This capability opens new avenues for studying the ultrafast motion of domain walls [32] or large skyrmions on nanometer length scales [33–35]." I would disagree with those statements. Domain walls might not be the most important objects to study with ultrafast techniques, particularly, if the microscopy cannot resolve domain walls at all. Second. what is a large skyrmions on a nanometer length scale? I suggest to rephrase this statement.

Reply: As has been demonstrated in the very recent pioneering ultrafast diffraction experiments by R. Jangid *et al.* (Phys. Rev. Lett. **131**, 256702 (2023)), at the frontier of current research it is generally keenly needed to observe and understand the ultrafast magnetic domain wall motion. Despite that the domain pattern retrieved in that specific work above with an average width of about 100 nm may not be fully resolved by our photoelectron microscopy (PEEM) in great detail, it has been shown in previous PEEM experiments that a limited spatial resolution of about 600 nm is fully sufficient to accurately determine the position of the domain wall with a 30 nm precision (Phys. Rev. B **81**, 224418 (2010)). In addition, here we use "large skyrmions" to differentiate them from the skyrmions having a size of less than 10 atomic distance (Phys. Rev. Lett. **114**, 177203 (2015); Nat. Commun. **11**, 5925 (2020)).

(A.12) Reviewer A: "In summary, I recommend a major revision. The authors also need to demonstrate why PRL is the appropriate journal, I feel that PRB would be the better choice for this work."

Reply: We thank the Reviewer for kindly providing us with the opportunity to revise. With our detailed answers above, we are confident that all the scientific questions are now satisfactorily addressed. Moreover, since our joint experimental and theoretical work is a major progress of photoelectron microscopy on magnetic imaging, covering a wide range of readership from surface science, magnetism, and synchrotron research communities, we hope that our choice to publish in the Physical Review Letters is reasonably justified.

Reviewer B

(B.1) Reviewer B: "In a revision, however, the authors should address the requirements for surface preparation and the need for crystalline samples. It would also be interesting to know how much the intensity decreases in the dark-field images shown."

Reply: The preparation of sample surfaces includes standard surface science procedures in the ultrahigh vacuum chamber such as cleaning with noble gas ion sputtering and annealing in ultrahigh vacuum to recover the surface crystallinity. The surface quality has been confirmed by low-energy electron diffraction as well as Auger electron spectroscopy and x-ray photoelectron spectroscopy. The crystalline order of the surfaces is needed for the well-defined momentum distribution of photoelectrons. We added a corresponding sentence on the second page.

The intensity decrease in our dark-field imaging condition is roughly within one order-of-magnitude due to the specific photoelectron angular distribution using the photon energy in our experiments. In general, dark-field imaging in photoelectron microscopy (PEEM) does not necessarily lead to a reduced intensity since it depends strongly on the electronic band structure and the associated photoemission matrix elements (Rev. Sci. Instrum. **83**, 083706 (2012)).